Alkenes & Alkynes

Ingyu Bahng

June 2026

Contents

1 Alkenes

Core Concept 1.1: Alkene Orbital Structure

The orbital structure of an alkene double bond arises from the combination of a σ bond and a π bond. This additional π overlap results in a shorter bond length relative to an alkane single bond: whereas a typical C–C single bond measures roughly 1.5 A, the C=C double bond is contracted to approximately 1.3 A.

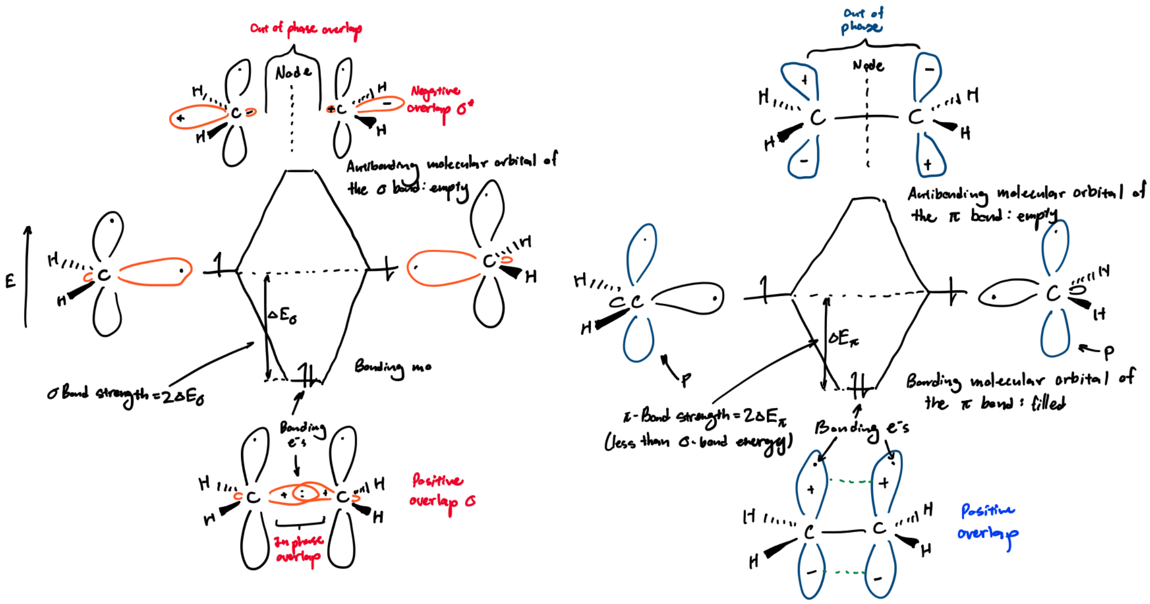


Figure 1: Alkene σ (*left*) and π (*right*) bonding orbitals

Core Concept 1.2: Alkene Bond Strength

The π bond of an alkene is relatively weaker than its partner σ bond because the antibonding π^* orbital sits closer in energy to the bonding π orbital than the antibonding σ^* orbital sits to the bonding σ orbital. As a result, breaking an alkene π bond requires roughly 65 $kcal\ mol^{-1}$, whereas breaking the σ bond requires 108 $kcal\ mol^{-1}$.

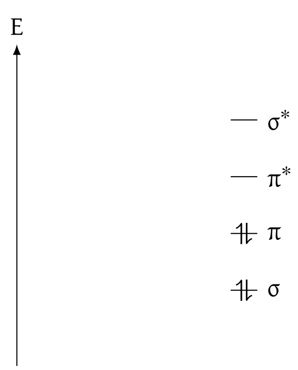


Figure 2: Molecular bonding orbitals levels of σ, π, π^*, and σ^* in an alkene double bond; *notice that the distance between π and π^* is less than that of σ and σ^**

By comparison, the typical alkane σ C–C bond strength is roughly 88 $kcal\ mol^{-1}$. The alkene σ bond is therefore stronger than the alkane σ bond, since the alkene σ bond is formed from sp^2 hybridized orbitals rather than the sp^3 orbitals used in alkanes. This **greater s-orbital character**—33% versus 25%, respectively—corresponds to greater bond strength, as the bonding e^-s are held closer to the nucleus relative to p orbitals.

Consequently, when comparing the pK_a values of the substituent hydrogens on alkenes and alkanes, alkenes exhibit lower pK_a values than alkanes.

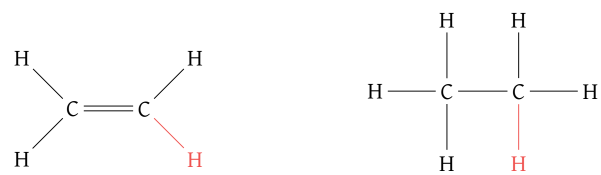


Figure 3: Ethene (*left*) vs ethane (*right*) pK_a values

In principle, then, the reaction between methyllithium and an alkene below will proceed to a product-favoring equilibrium, since the alkene is more acidic (*better able to release a proton*) than the alkane.

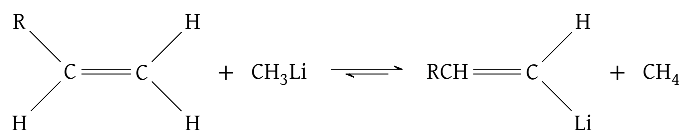


Figure 4: Methyllithium reaction showcasing alkene favoring equilibrium due to pK_a differences

In practice, however, the reaction shown in Figure 4 faces regio- and stereoselectivity issues, since the remaining substituent hydrogens may dissociate equally or more favorably than the one depicted. To overcome these problems, chemists instead employ the formation of **organometallics** (*Core Concept 2.3*).

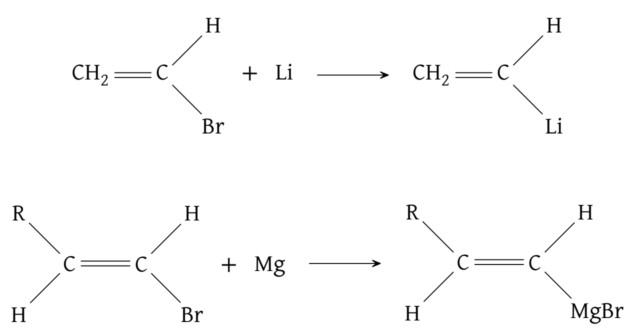


Figure 5: Methyllithium reaction showcasing alkene favoring equilibrium due to pK_a differences

Core Concept 1.3: Relative Stability of Alkenes

Consider the **heat of hydrogenation ($\Delta H_{hydrogenation}$)** of the following butene isomers upon conversion to butane.

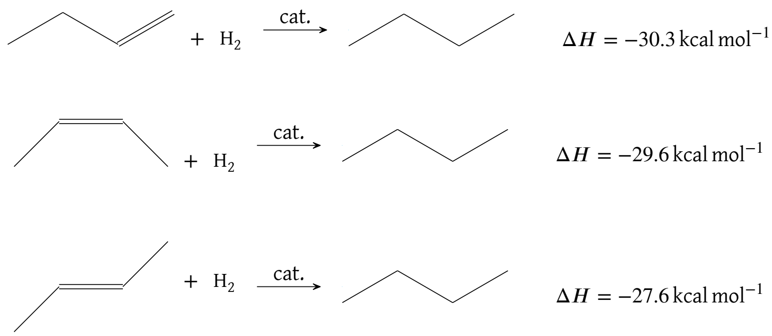


Figure 6: Butene isomer conversions to butane

The greater relative stability of internal double bonds over terminal ones, and of trans over cis substituents, arises from two factors.

(a) **Hyperconjugation** allows the temporary charge imbalance arising in the transition state to be better distributed among neighboring orbitals in the internal bond butene than in its terminal counterpart.

(b) **Steric hindrance** between the cis methyl groups increases the relative instability of the cis butene compared to its trans counterpart.

Figure 7: Steric hindrance between methyl groups of cis (*left*) vs trans (*right*) butene; *more steric hindrance exists between methyl groups of cis butene because of closer methyl group proximity*

The general order of alkene stability thus follows as shown below.

$$\mathrm{CH_2{=}CH_2} < \mathrm{RCH{=}CH_2} < \mathrm{RCH{=}CHR_{cis}} < \mathrm{RCH{=}CHR_{trans}} < \mathrm{RCH{=}CR_2} < \mathrm{R_2C{=}CR_2}$$

1.1 Alkene Synthesis

Core Concept 1.4: Elimination & Saytzev Rule

Synthesizing alkenes from alkanes through E2 elimination requires the consideration of several factors.

(a) **Base Size**

Consider an E2 reaction **with a small base** on the following RX alkane.

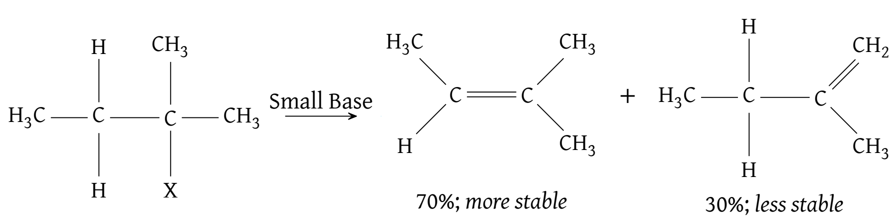


Figure 8: E2 reaction on a haloalkane using a small base

Notice that the internal double bond alkene is favored over the terminal version, since the advantages of hyperconjugation outweigh the minimal steric hindrance associated with a small base. In contrast, an E2 reaction **using a large base** (*Figure 9*) favors the terminal version, since steric hindrance exerts a comparatively larger influence on transition state stability, sufficient to override the hyperconjugative favorability of the internal double bond product.

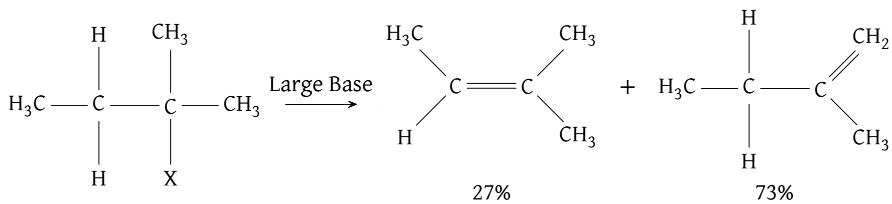


Figure 9: E2 reaction on a haloalkane using a large base

This phenomenon is termed the **Saytzev Rule**, which states that non-bulky bases lead to the more stable internal double bond product. The corresponding energy diagrams of the two reactions are shown below to depict the energy levels of the reaction participants throughout the reaction.

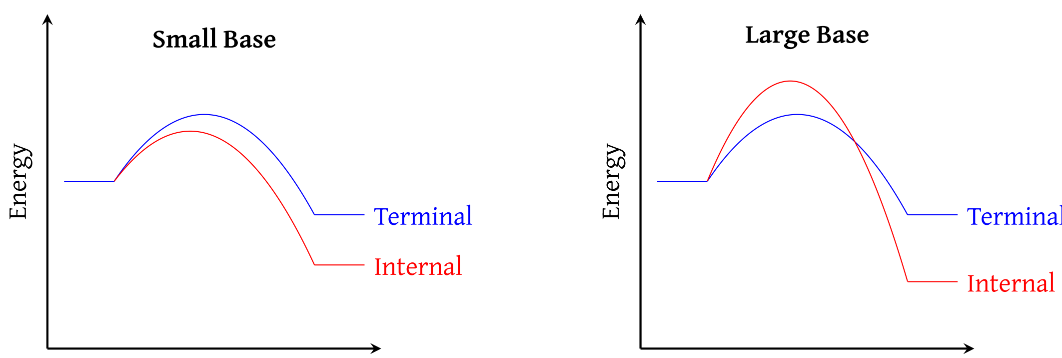


Figure 10: Energy diagrams of haloalkane elimination reactions producing alkenes; [Small Base] *lower energy transition state (red) favors formation of more stable product*; [Large Base] *with a large base, the lower energy transition state (blue) now favors formation of less stable product*

(b) **Stereoselectivity**

The presence of **multiple stereoisomers**—particularly among the internal double bond alkenes formed via E2 reactions with small bases—is another factor a chemist must consider. Although the E2 reaction is stereoselective, favoring the more stable trans isomers over the sterically hindered cis isomers, this preference is not absolute (*some cis isomers do form*).

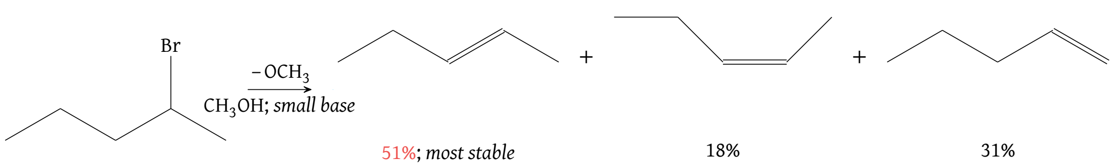


Figure 11: Alkene product mixture from E2 elimination of 2-bromo-pentane

(c) **Other Factors**

Other factors, such as preferable precursor conformations—namely rotamers—play a role as well.

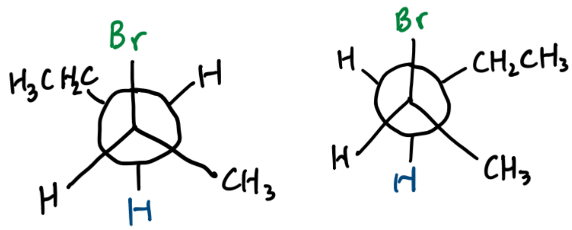


Figure 12: Precursor conformations of 2-bromo-pentane; *the left rotamer conformation is more favorable because the ethyl and methyl substituents are anti to one another whereas the right conformation is unfavorable due to the steric hindrance caused by the gauche positioning of the ethyl and methyl groups*

Keep in mind that elimination is also **stereospecific** as shown in the following figure.

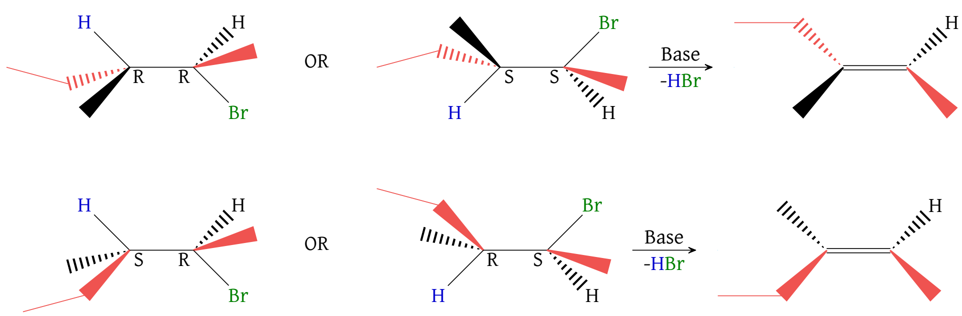


Figure 13: Stereospecificity of HX elimination

Core Concept 1.5: Alkenes from ROH By Dehydration

Alkenes can also be synthesized by exposing alcohols to acids and subsequently removing the resulting H_2O byproduct through dehydration.

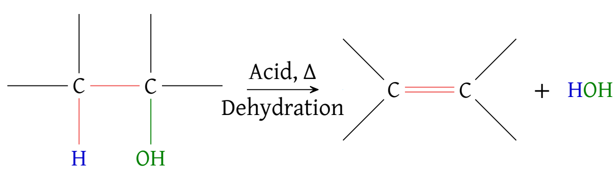


Figure 14: Formation of an alkene via alcohol dehydration

The mechanism behind the reaction—using sulfuric acid and ethanol as an example—is as follows.

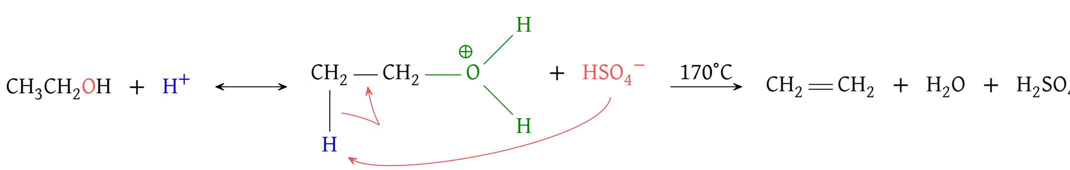


Figure 15: Alcohol dehydration example using H_2SO_4 and ethanol; HSO_4^- unusually acts as a base in this scenario

Due to the effects of steric hindrance and hyperconjugation, the use of E1 versus E2 and the resulting stereoisomer product ratio differ across $R_{primary}$, $R_{secondary}$, and $R_{tertiary}$.

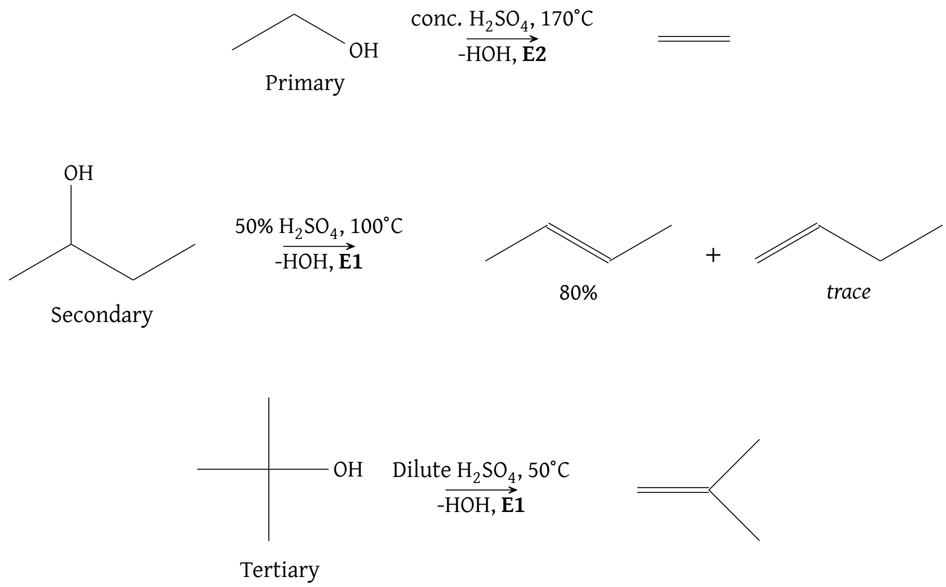


Figure 16: Alkene product ratios from alcohol dehydration across $R_{primary}$, $R_{secondary}$, and $R_{tertiary}$

A major limitation of this approach is the formation of a **carbocation transition state**, which opens the door to **unintended rearrangement pathways** such as **hydride** (*Core Concept 3.1*) or **alkyl shifts** (*Core Concept 3.3*) where possible. As a result, this alkene synthesis pathway has limited utility in practice.

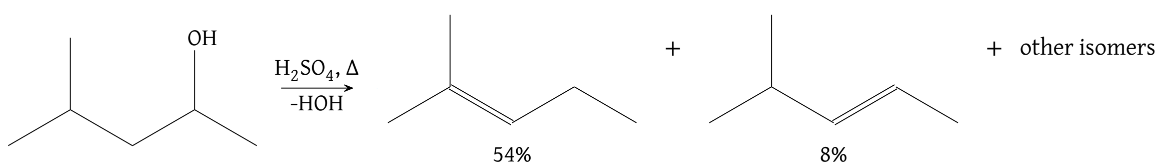


Figure 17: Product mixtures from alcohol dehydration due to hydride shifts

1.2 Alkene Addition Reactions

Core Concept 1.6: Alkene Addition Reactions

Alkenes serve as fundamental building blocks for organic synthesis and the preparation of polymer materials. Their defining feature, the **unsaturated π bond**, reacts through **addition reactions**, in which the π bond is broken and a diatomic or compound molecule is added across the relevant carbons of the former alkene.

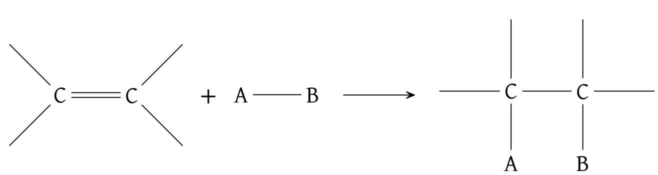


Figure 18: Outline alkene addition reaction; *this reaction is usually exothermic*

Generally speaking, alkene addition reactions exhibit $\Delta H^{\circ} < 0$; however, because the reaction is a combination reaction in which two reactant molecules form a single product molecule, $\Delta S^{\circ} < 0$, rendering it **entropically unfavorable**. In most cases, the enthalpic favorability outweighs the magnitude of entropic unfavorability. In some reactions, however, or at sufficiently high temperatures, ΔG° becomes positive, rendering these addition reactions thermodynamically unfavorable.

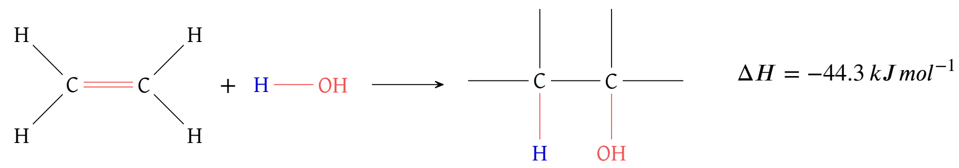


Figure 19: Alkene hydration reaction; *the reaction is enthalpically favorable ($\Delta H^{\circ} = -44.3\ kJ\ mol^{-1}$) but entropically unfavorable in standard conditions, thus the Gibbs Free Energy is much lower ($\Delta G^{\circ} = -6.1\ kJ\ mol^{-1}$)*

1.2.1 Catalytic Hydrogenation

Core Concept 1.7: Catalytic Hydrogenation

Catalytic hydrogenation refers to the addition of H_2 to an alkene, effectively removing the alkene double bond and converting the target molecule into an alkane if that was the only double bond present. On its own, this process would require enormous pressure and extreme temperatures to proceed; however, **catalysts**—such as Pd/C or PtO_2—enable an alternative mechanism with a lower E_a that chemists frequently employ.

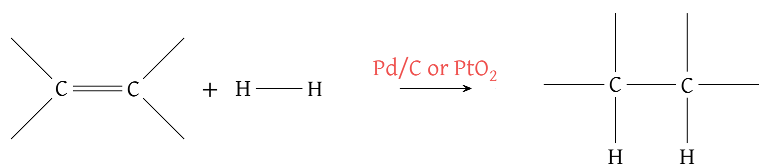


Figure 20: Catalytic hydrogenation reaction outline

It is also important to note that the hydrogens are added **stereospecifically** to the **same side (syn)** of the target double bond. As a result, the resulting product mixture is **racemic**.

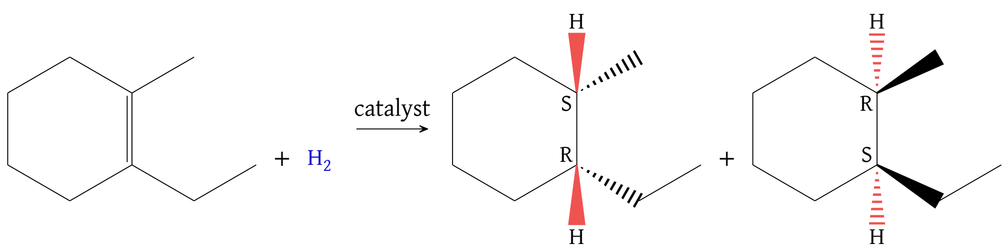


Figure 21: Stereospecific catalytic hydrogenation of asymmetrical cyclohexane leading to racemic product mixture; *notice that the hydrogen addition is always syn*

1.2.2 Electrophilic Additions

Core Concept 1.8: Electrophilic Additions

Because the π bond is **e^- rich**, it readily **reacts with a given polar reagent $A^{\delta+}$–$B^{\delta-}$**, in which the partially positive end $A^{\delta+}$ is drawn toward the e^--rich double bond.

In the case where $A^{\delta+}$ is H^+, the ensuing reaction is simply the reverse of an E1.

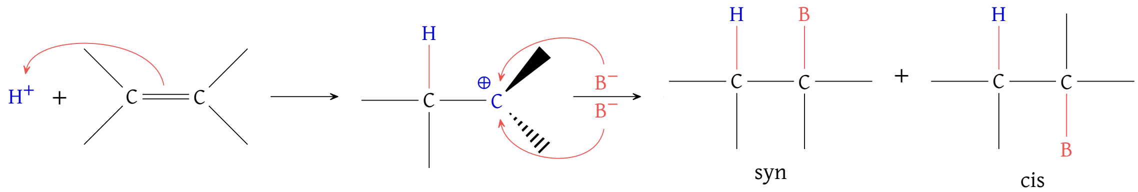


Figure 22: Outline reaction of alkene electrophilic addition; *mind that the syn and cis versions have two possible arrangement as shown by the absence of dashed and wedged bonds*

Core Concept 1.9: Hydrohalogenation

Hydrohalogenation is a type of electrophilic addition in which **HX is added** to the alkene double bond, where X is a halogen.

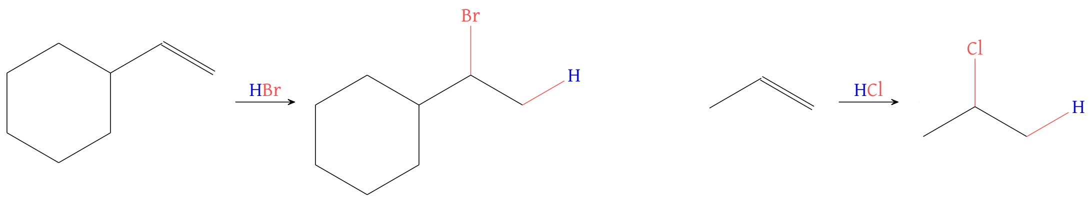


Figure 23: Hydrobromination (*left*) and hydrochlorination (*right*)

Notice that H^+ adds to the **less substituted** carbon, since doing so results in a more substituted carbocation (*a more stable transition state*) at the other carbon, favoring carbocation formation in the order tertiary > secondary > primary. This phenomenon is called **Markovnikov's Rule**, which states that among the possible stereospecific pathways, the one proceeding through the most substituted carbocation has the lowest-energy transition state and is therefore strongly favored over the alternatives.

Core Concept 1.10: Markovnikov Hydration

Markovnikov hydration refers to the conversion of an alkene to an alcohol through the **addition of H_2O** in the presence of an acid catalyst—the reverse of the alkene synthesis reaction covered in Core Concept 1.5.

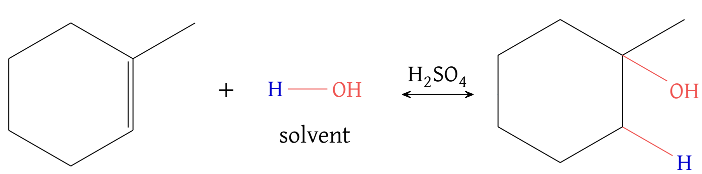


Figure 24: Markovnikov hydration reaction on asymmetrical cyclohexene

The underlying mechanism of the reaction is shown below. Notice that H^+ is regenerated by the end of the reaction.

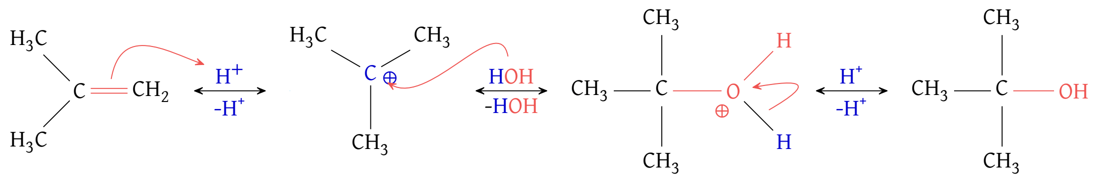


Figure 25: Markovnikov hydration mechanism; *H^+ is consumed and regenerated at the start and end of the reaction process*

A mixture of alkenes, transient intermediates, and alcohols is nonetheless bound to form, since Markovnikov hydration exists in dynamic equilibrium with the pathway previously covered in Core Concept 1.5. Because the temporary presence of a carbocation introduces hydride shifts, catalytic H^+ converts a single alkene species into a mixture of stereoisomeric alkenes, with product ratios determined by relative thermodynamic stability, governed primarily by steric hindrance and hyperconjugation.

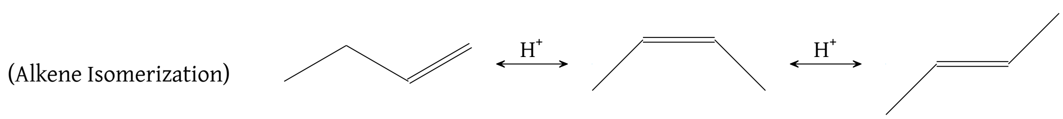


Figure 26: Equilibrium product concentrations reflecting relative thermodynamic stabilities (*bottom*) due to alkene isomerization from catalytic H^+ with a nonnucleophilic counter ion (*top*)

The complete equilibrium mechanism between an alcohol and its corresponding alkene products is shown in the following figure, using butanol/butene as an example.

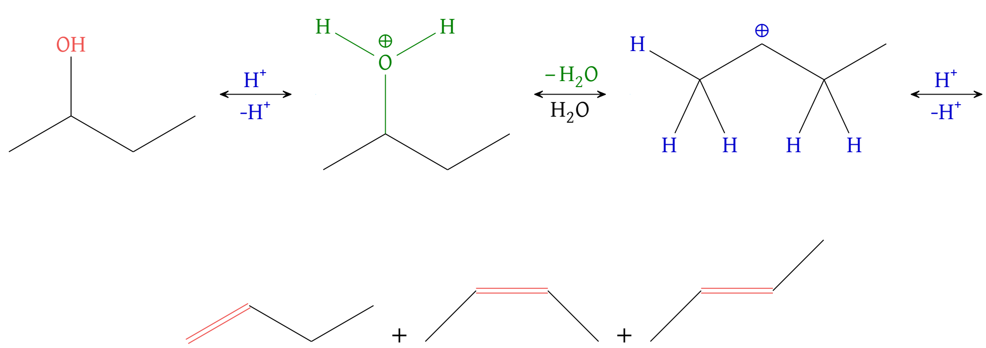


Figure 27: Equilibrium of the Markovnikov hydration reaction; *fast deprotonation-reprotonation-deprotonation leads to rapid equilibration of three alkene products*

Core Concept 1.11: Halogenation

Halogenation is the addition of a diatomic halogen molecule X_2 to an alkene double bond. In contrast to the addition of HX (*Core Concept 1.9*), which is inherently polarized due to electronegativity differences, X_2 is not inherently polarized but rather **becomes polarized as it approaches the e^- rich alkene double bond.**

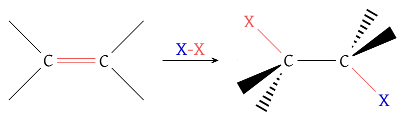


Figure 28: Outline alkene halogenation reaction; *notice that X_2 gets polarized during approach to alkene*

The underlying mechanism is shown in the figure below.

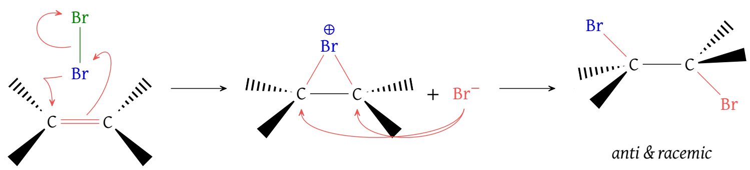


Figure 29: Halogenation (*bromination*) of alkenes mechanism

Notice the formation of a halonium ring, since the alternative carbocation state would otherwise bear the full burden of its positive charge. Because the second attack on the bridged halonium ion occurs exclusively at the backside of the first halogen attack, halogenation is **stereospecifically anti (not syn)**. Likewise, if the second attack is thermodynamically equally likely to occur at either backside carbon—meaning the initial alkene was perfectly **symmetrical**—the resulting products form a **racemic mixture**.

If the initial alkene is **asymmetrical**, however, the halonium ion intermediate becomes highly distorted, such that the bond between the halogen and the more substituted carbon is **longer**. Likewise, as the halogen draws more e^- density toward itself, the more substituted carbon develops a greater δ^+ due to hyperconjugation. As a result, the halonium ring transition state **already exhibits a high degree of carbocation** character and thus more closely resembles S_N1 type behavior, even though the backside attack is S_N2 by definition due to its exclusively anti outcome. Consequently, the backside nucleophilic attack occurs at the more substituted carbon, an interaction in which the influence of charge outweighs steric hindrance in determining regioselectivity.

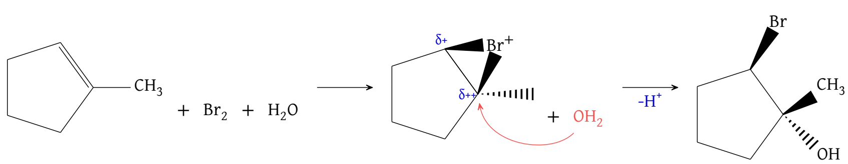


Figure 30: Backside attack of halonium ring ion by H_2O; *notice that the more substituted carbon possess a greater partial positive charge relative to the adjacent carbon*

Separately, the intermediate halonium ring ions can also be intercepted by other nucleophiles, such as HOH (*Figure 30*) or ROH (*Figure 31*).

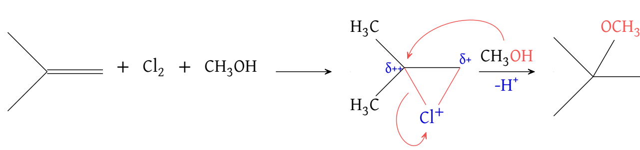


Figure 31: Backside attack of halonium ring ion by ROH; *this reaction forms haloethers*

Core Concept 1.12: Oxymercuration–Demercuration

Oxymercuration–demercuration is effectively equivalent to the **Markovnikov hydration** process (*Core Concept 1.10*), in which H_2O is added to remove an alkene double bond. However, oxymercuration–demercuration **avoids carbocation** formation, which prevents unintended rearranged alcohol products and yields a purer product mixture.

The process employs mercury(II) acetate and sodium borohydride in a two-step sequence to achieve H_2O addition.

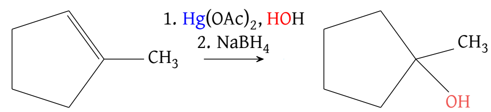


Figure 32: Oxymercuration–demercuration chemical reaction

(1) **Oxymercuration**

Similar to the **alkene halogenation** process (*Core Concept 1.11*), the large mercury atom forms a bridged, unevenly distributed mercurinium ion rather than a full carbocation, forcing the subsequent H_2O attack to occur strictly at the more substituted carbon.

(2) **Demercuration**

Afterward, $NaBH_4$ acts as a reducing agent that cleaves the C–Hg bond and replaces it with a hydrogen.

Likewise, the use of ROH allows for the addition of an –OR group, just as H_2O adds a –OH group.

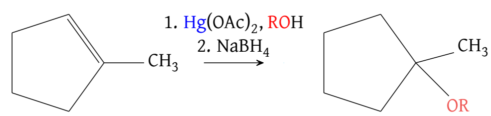
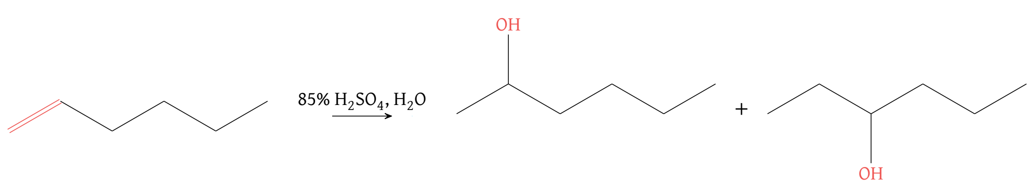
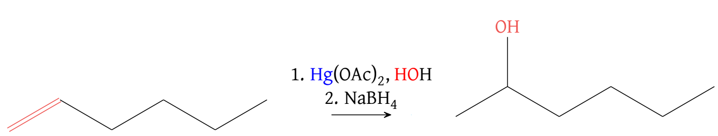


Figure 33: Oxymercuration–demercuration chemical reaction using ROH

When compared to Markovnikov hydration (*Core Concept 1.10*), oxymercuration–demercuration is the preferred method preferred for converting an alkene to an alcohol, owing to the purity of the resulting product mixture.



70%; 2-Hexanol (*normal*) 30%; 3-Hexanol (*rearranged*)



95%

Figure 34: Markovnikov hydration (*top*) vs oxymercuration–demercuration (*bottom*); *the latter method allows for a much purer product mixture compared to the former*

1.2.3 Hydroboration-Oxidations

Core Concept 1.13: Hydroboration-Oxidations

Hydroboration-oxidation is a two-step method by which chemists add a molecule of H_2O across the double bond of an alkene, with the hydroxyl group installed at the **less substituted carbon** (*recall that Markovnikov hydration and oxymercuration–demercuration instead deliver H_2O to the more substituted carbon*). In this process, BH_3 reacts regioselectively with three equivalents of alkene, and the resulting trialkylborane (BR_3) is subsequently oxidized and cleaved to yield three equivalents of ROH.

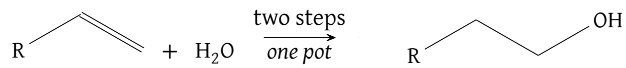


Figure 35: Outline reaction of hydroboration-oxidation

(1) **Hydroboration**

When treated with BH_3, three equivalents of alkene bond to the borane such that the **less substituted carbon attaches to boron**, a regiochemical outcome that arises from two major factors.

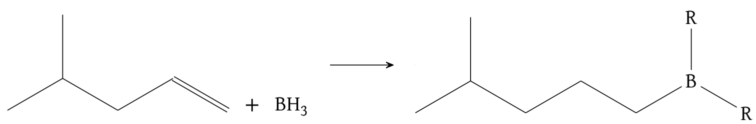


Figure 36: Hydroboration of terminal alkene; *mind that the R groups in the product is the same is the one fully displayed alkyl substituent*; *notice that the boron attaches to the less hindered end of the terminal alkene double bond*

- **Transition State Stabilization**

  As the e⁻ rich π bond of the alkene attacks the empty, unhybridized p orbital of boron, the two alkene carbons do not share the resulting electronic burden equally. The carbon **not** forming the C–B bond progressively loses electron density, developing a partial positive charge (δ^+) in the transition state. Consequently, when the less substituted carbon forms the C–B bond, the more substituted carbon is left to stabilize this developing charge through hyperconjugation, yielding a more stable transition state than the alternative regiochemical outcome.

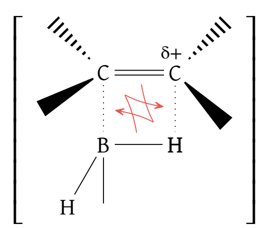


Figure 37: Hydroboration transition state

- **Bond Polarization**

  A second, more subtle factor is the inherent electronegativity difference between boron ($\approx$ 2.0) and hydrogen ($\approx$ 2.1), which polarizes the B–H bond such that boron carries a slight positive charge (δ^+) and hydrogen a slight negative charge (δ^-). Consequently, the slightly electropositive boron preferentially pairs with the e⁻ rich attack from the less substituted alkene carbon, while the slightly electronegative hydrogen preferentially pairs with the developing partial positive charge on the more substituted alkene carbon.

Another defining feature of hydroboration is the **syn** stereospecificity of the resulting alkylborane. Because the boron and hydrogen attacked by the alkene's π orbital are physically bonded to one another, they are delivered to the double bond simultaneously, constraining them to add to the same face of the alkene.

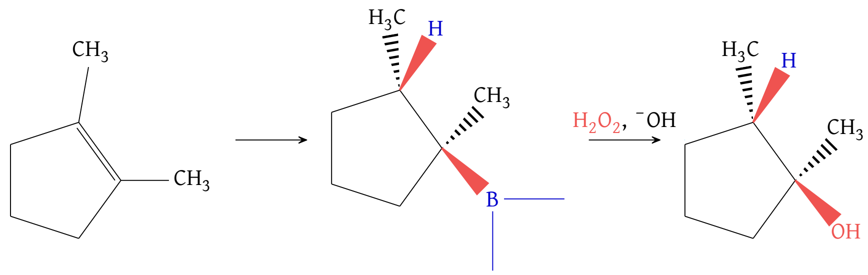


Figure 38: Syn addition of B–H

(2) **Oxidation**

The resulting alkylborane (R_3B) is then oxidized by a solution of H_2O_2 and ^-OH, yielding a trialkyl borate ($(RO)_3B$) via the alkyl-shift mechanism shown below.

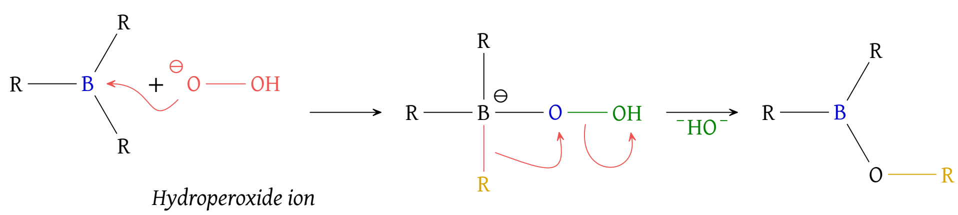


Figure 39: Oxidization of BR_3

Finally, base-assisted hydrolysis of $(RO)_3B$ by H_2O furnishes the three desired ROH molecules, with a borate ion generated as a byproduct.

$$(RO)_3B + 3\,NaOH \xrightarrow{H_2O} Na_3BO_3 + 3\,ROH$$

This transformation is often termed **anti-Markovnikov hydration**, since it adds a molecule of H_2O across the alkene but places the –OH group on the less (*rather than more*) substituted π-bond carbon.

1.2.4 Electrophilic Carbene Additions

Core Concept 1.14: Carbene Synthesis

Carbenes (:CR_2) are sp^2-hybridized, **six-electron** species bearing a lone pair and an empty p orbital; their parent compound is methylene (:CH_2). Conceptually, carbenes can be regarded as the products of **deprotonating carbocations**.

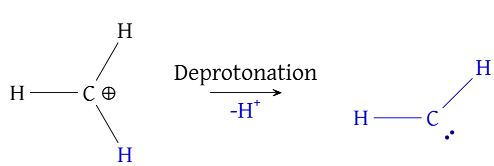


Figure 40: Deprotonation of carbocation

Owing to their electron deficiency, carbenes are extremely unstable, short-lived intermediates that behave as powerful **electrophiles**. Below are the three primary pathways chemists use to generate these intermediates in the laboratory.

(a) **Methylene from diazomethane (CH_2N_2)**

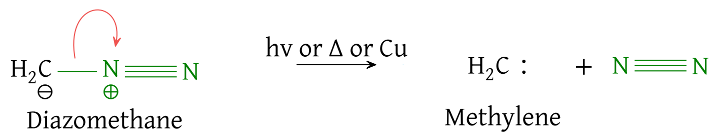


Figure 41: Methylene from diazomethane

Note that this reaction is essentially **irreversible**: the exceptional stability of the resulting N_2 drives the reaction forward, cleanly liberating highly reactive methylene.

(b) **Dichlorocarbene from chloroform**

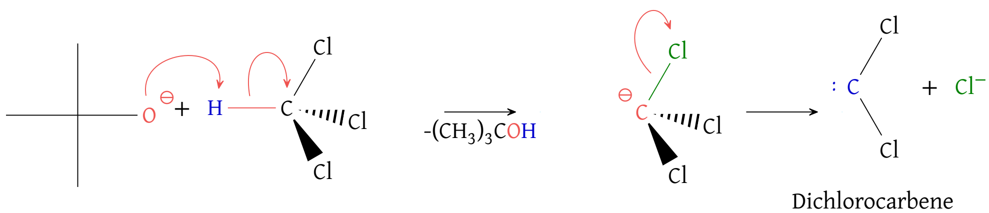


Figure 42: Dichlorocarbene from chloroform

This reaction proceeds via α-elimination (*distinct from the standard β-eliminations, E1 or E2, typically used to form alkenes*), driven by the strong base $(CH_3)_3CO^-$ and the departure of a highly electronegative chlorine atom.

(c) **Simmons-Smith reagent**

As mentioned previously, free methylene is so reactive that it can be hazardous to handle directly. The Simmons–Smith reaction offers a much safer, more controlled alternative.

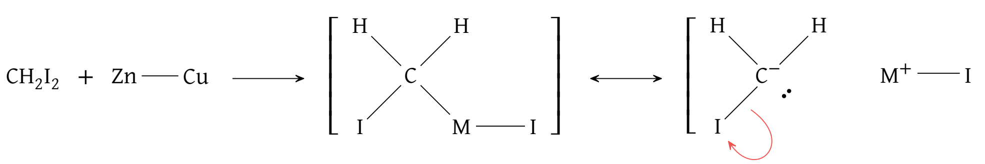


Figure 43: Simmons-Smith reagent methylene formation

Treatment of diiodomethane with a zinc–copper couple generates an organometallic species that behaves as a hybrid between a stable molecule and a free methylene.

Core Concept 1.15: Electrophilic Carbene Additions

Given their inherent reactivity, carbenes add to alkenes in a **syn stereospecific manner** via the general mechanism shown below, forming a cyclopropane ring as the product.

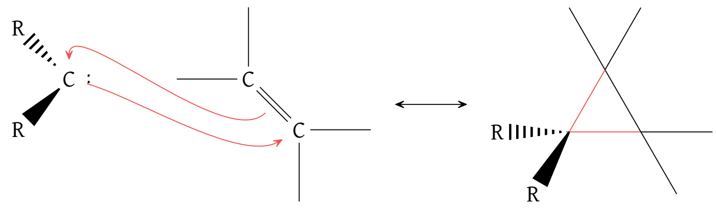


Figure 44: Outline reaction of electrophilic carbene additions to alkenes

In practice, because carbene species are transient, carbene addition must occur in situ, concurrent with one of the carbene synthesis processes described above (*Core Concept 1.14*).

(a) **Methylene from diazomethane (CH_2N_2)**

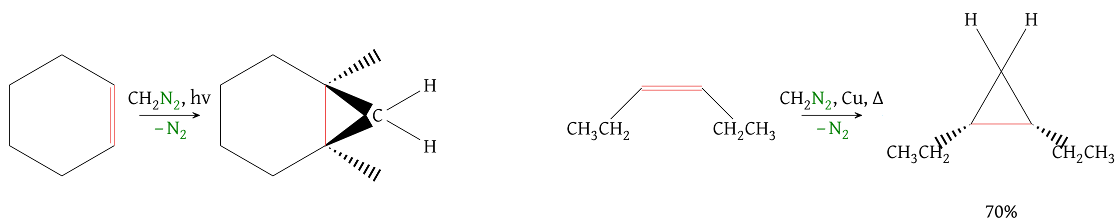


Figure 45: Examples of methylene formation from diazomethane occurring simultaneously with its addition to an alkene

(b) **Dichlorocarbene from chloroform**

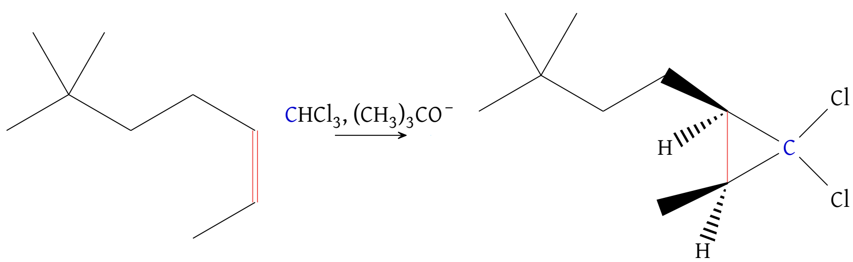


Figure 46: Dichlorocarbene formation from chloroform occurring simultaneously with its addition to an alkene; *mind that this reaction is always in syn configuration*

(c) **Simmons-Smith reagent**

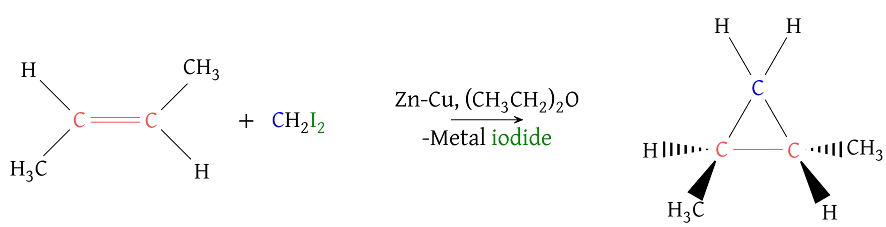


Figure 47: Simmons-Smith reagent formation occurring simultaneously with its addition to an alkene

1.2.5 Electrophile Oxidation

Core Concept 1.16: Peroxycarboxylic Acids

Peroxycarboxylic acids are a class of acids possessing the general structure shown below.

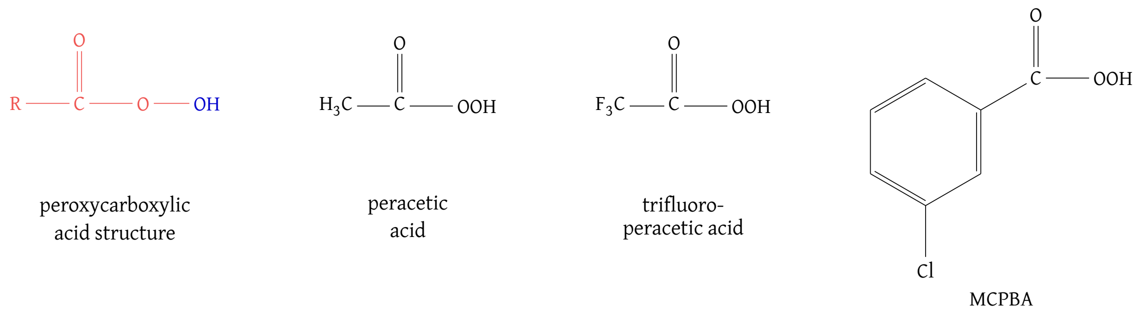


Figure 48: General peroxycarboxylic acid structure (*left*) and commonly used peroxycarboxylic acids

Treatment of alkenes with these compounds results in the **syn addition of an oxygen atom**, forming the cyclic ether product outlined below.

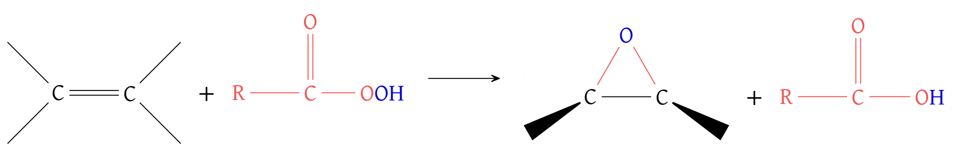


Figure 49: Peroxycarboxylic acid oxidation of an alkene

The underlying mechanism, which is **concerted**, proceeds as follows.

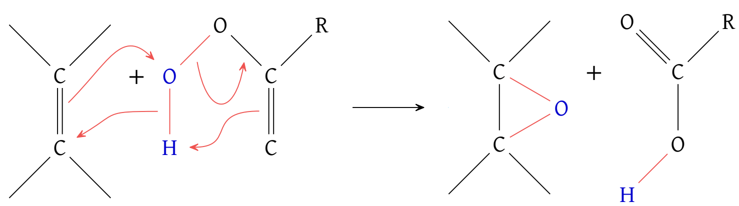


Figure 50: Mechanism of peroxycarboxylic acid addition to alkenes

It is also important to note that the **rate of oxacyclopropanation increases with alkyl substitution**, since greater hyperconjugation allows for a greater degree of e^- donation from the alkene's π orbital.

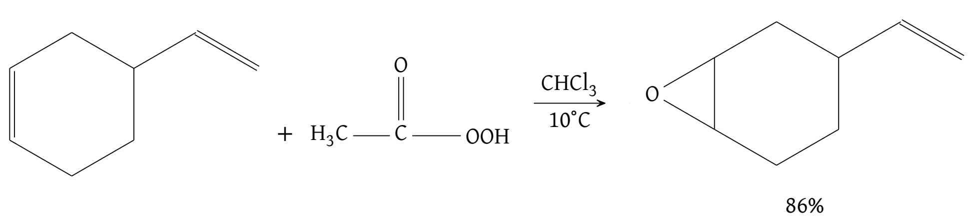


Figure 51: Rate of oxacyclopropanation increases with alkyl substitution, since hyperconjugation allows for greater e^- donation; *notice that the more favorable product derives from the oxacyclopropanation of the more substituted alkene double bond*

In practice, peroxycarboxylic acid addition usually serves as a preparatory step for further transformations. Two common applications are shown below.

(a) **Double Nucleophile Syn-Addition**

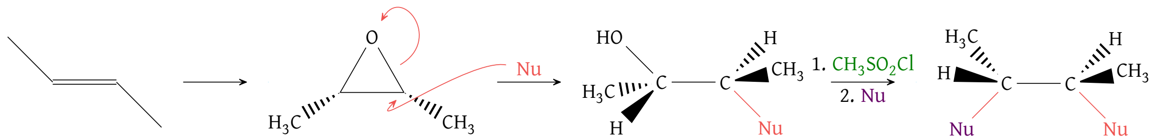


Figure 52: Double nucleophilic syn addition to alkene double bond; *CH_3SO_2Cl turns the OH group into a sulfonate leaving group*

(b) **Anti Stereospecific Dihydroxylation**

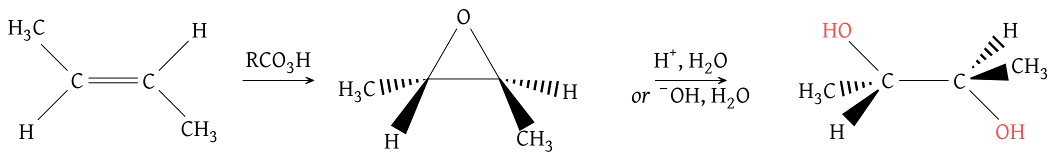


Figure 53: Anti-dihydroxylation of alkenes through oxacyclopropanation; *notice that the end product is meso, meaning it has two or more chiral centers*

Core Concept 1.17: OsO_4 Syn-Dihydroxylation

Treating alkenes with OsO_4, followed by reduction with an agent such as H_2S, results in syn-dihydroxylation of the alkene double bond.

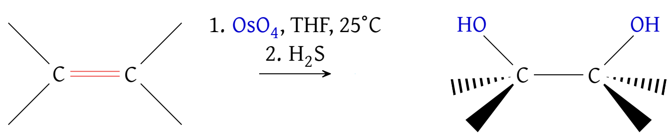


Figure 54: Syn-dihydroxylation of an alkene using OsO_4

The underlying mechanism proceeds as follows, forming an osmate ester intermediate prior to reduction in the H_2S step.

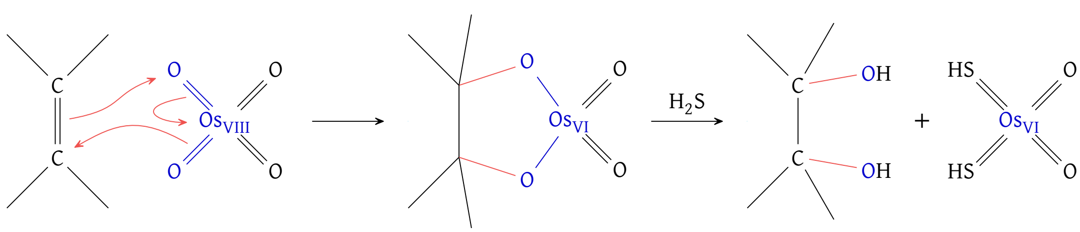
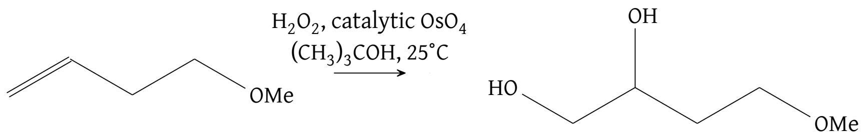


Figure 55: Mechanism of syn-dihydroxylation using OsO_4

The osmium compound consumed in this process can be reoxidized by the addition of H_2O_2, rendering it **catalytic** (*i.e., not net consumed until the H_2O_2 supply is depleted*). This is especially advantageous given the high cost of osmium.



4-Methoxy-1-butene 77%; *4-Methoxy-1,2-butanediol*

Figure 56: Catalytic OsO_4 through H_2O_2

Core Concept 1.18: Ozonolysis

Ozonolysis employs ozone (O_3)—a highly unstable, rapidly decaying species—together with a reducing agent to achieve complete oxidative cleavage of an alkene double bond.

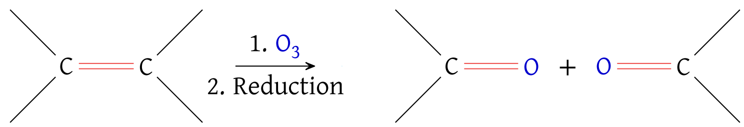
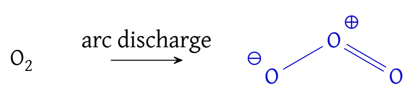


Figure 57: Ozonolysis reaction outline

Because ozone is such a transient species, it is not naturally available in sufficient quantities for laboratory use. To overcome this limitation, chemists pass O_2 through an **arc discharge**—a continuous, highly intense electrical breakdown of the gas—to generate a fresh stream of 2–4% O_3 gas safely in the lab.



2-4% in O_2

Figure 58: Laboratory formation process of ozone through arc discharge

Listed below are three of the most common reducing agents used for this purpose.

(a) **Zinc and Acid (Zn / CH_3COOH)**

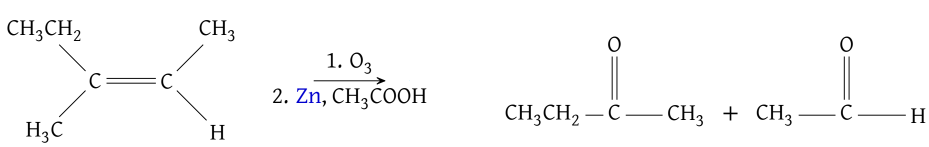


Figure 59: Ozonolysis reduction through zinc and acid

(b) **Catalytic Hydrogenation (H_2 / Pt)**

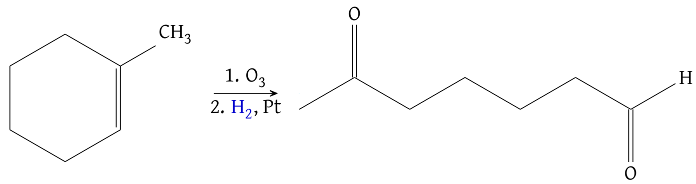


Figure 60: Ozonolysis reduction through catalytic hydrogenation; *this reaction yields a cycloalkene ring opening*

(c) **Dimethyl Sulfide (DMS; $(CH_3)_2S$)**

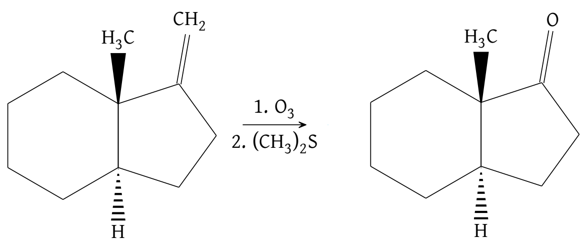


Figure 61: Ozonolysis reduction through dimethyl sulfide (DMS); *this reaction yields a "clipped off" carbon atom*

1.2.6 Radical Hydrobromination

Core Concept 1.19: Radical Hydrobromination

Radical hydrobromination refers to the peroxide-assisted (RO – OR) addition of HBr across an alkene double bond in an **anti-Markovnikov** manner, such that bromine attaches to the **less substituted carbon**. Recall that standard **hydrohalogenation** (*Core Concept 1.9*) instead follows Markovnikov regiochemistry, in which the halogen attaches to the more substituted carbon.

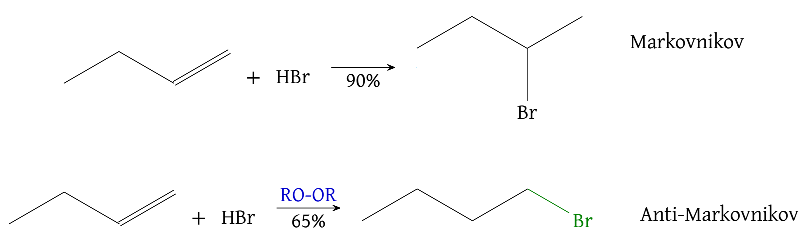


Figure 62: Markovnikov (*top*) vs anti-Markovnikov (*bottom*) HBr addition to alkene double bond; *the radical anti-Markovnikov version is much faster*

The underlying mechanism, initiated by peroxide radical initiators (*the O–O bond is weak, at approximately* 39 *kcal mol*$^{-1}$), is shown in the following figure.

(a) **Initiation Steps**

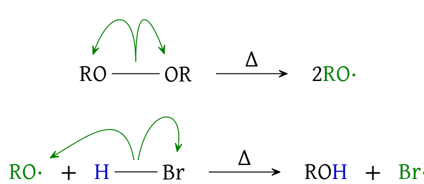
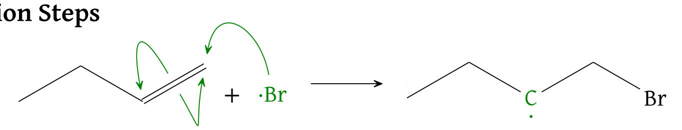


Figure 63: Initiation steps of radical hydrobromination; *the formation of the radical Br· begins the reaction*

(b) **Propagation Steps**



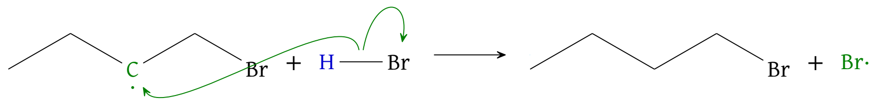


Figure 64: Propagation steps of radical hydrobromination; *in regular Markovnikov, H^+ is added to the less substituted radical and then Br^-; here, Br· is added and then H·; notice that the Br· is regenerated at the end to enter the next propagation cycle*

Notice that Br· adds to the less substituted carbon, leaving the resulting carbon radical on the more substituted carbon, where it is better stabilized by hyperconjugation than in the alternative regiochemical outcome.

This radical mechanism is not viable for HCl or HI, because one of the propagation steps becomes **endothermic** owing to the distinct electronegativities of Cl and I. As a result, the radical pathway is too slow, and the ionic, Markovnikov mechanism (*attachment to the more substituted carbon*) predominates instead. Bromine, by contrast, occupies the thermodynamic sweet spot between chlorine and iodine. Anti-Markovnikov addition of RSH proceeds by the same radical mechanism as HBr, similarly occupying that thermodynamic sweet spot.

2 Alkynes

Core Concept 2.1: Alkyne Orbital Structure

An alkyne triple bond consists of two mutually perpendicular π bonds together with a single sp hybridized σ bond.

The additional π overlap over an alkene orbital structure (*Core Concept 1.1*) results in an even shorter bond length: whereas the previously covered alkene double bond measures roughly as 1.3 *A*, the alkyne triple bond measures at roughly 1.2 *A*.

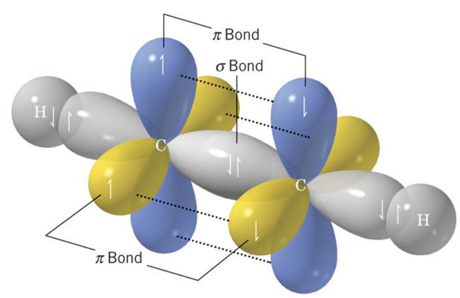


Figure 65: Alkyne triple bond orbital structure

Core Concept 2.2: Triple Bond Strength

Combustion of ethyne (HC ≡ CH) releases a substantial −311 kcal mol^{-1} of heat. Compared to ethane (CH_3CH_3), which releases −373 kcal mol^{-1}, ethyne releases proportionally more energy per carbon, since none of that energy must first be spent breaking the four additional C–H bonds present in ethane.

$$HC\equiv CH + 2.5\,O_2 \longrightarrow 2\,CO_2 + H_2O \qquad \Delta H^\circ = -311\,\text{kcal mol}^{-1}$$

$$CH_3CH_3 + 3.5\,O_2 \longrightarrow 2\,CO_2 + 3\,H_2O \qquad \Delta H^\circ = -373\,\text{kcal mol}^{-1}$$

Because alkyne π bonds exist under a higher degree of strain, complete hydrogenation of both alkyne π bonds releases more energy than hydrogenation of two analogous alkene π bonds would, which is roughly −60 kcal mol^{-1}. Notice that both terminal and internal hydrogenation have $\Delta H^\circ < -60\,\text{kcal mol}^{-1}$.

$$(\textit{terminal})\ CH_3CH_2C\equiv CH + 2\,H_2 \xrightarrow{\text{catalyst}} CH_3CH_2CH_2CH_3 \qquad \Delta H^\circ = -69.9\,\text{kcal mol}^{-1}$$

$$(\textit{internal})\ CH_3C\equiv CCH_3 + 2\,H_2 \xrightarrow{\text{catalyst}} CH_3CH_2CH_2CH_3 \qquad \Delta H^\circ = -65.1\,\text{kcal mol}^{-1}$$

Note the difference in net energy release: the internal triple bond is more stable than the terminal one owing to greater hyperconjugative stabilization, thus the internal scenario consumes more energy in order to break the same bonds, hence the lesser energy release compared to the terminal scenario.

Core Concept 2.3: Relative Acidity of Alkynes

In typical hydrocarbons (*alkanes and alkenes*), the C–H bond is considered essentially non-acidic. Terminal alkyne C–H bonds, however, defy this generalization: when treated with a sufficiently strong base, the terminal H^+ can be removed.

$$R-C\equiv C-H + B^- \longrightarrow R-C\equiv C^{\ominus} + HB \qquad pK_a = 25$$

Figure 66: Alkyne terminal proton donation in the presence of a sufficiently strong base

$$\underset{pK_a\approx 50}{H_3C{-}CH_3} \quad < \quad \underset{pK_a\approx 44}{H_2C{=}CH_2} \quad < \quad \underset{pK_a\approx 25}{HC{\equiv}CH}$$

This acidic trend is explained by the **greater s character of the relevant hybrid orbital**. Because an s orbital is spherical and lies much closer to the positively charged carbon nucleus than a directional, elongated p orbital, the lone pair of electrons left behind after deprotonation is held very close to that nucleus. This electrostatic proximity stabilizes the negative charge substantially, making the corresponding alkynyl conjugate base comparatively more stable. Chemists exploit this acidity in the following example applications.

(a) **Sodium Amide ($NaNH_2$)**

$$H-C\equiv C-H \xrightleftharpoons[-NH_3]{NaNH_2} H-C\equiv C^- \ Na^+ \xrightleftharpoons[-NH_3]{NaNH_2} Na^+\ C^-\equiv C^-\ Na^+$$

Figure 67: Alkyne acidity and sodium amide

(b) **Organolithium Reagents**

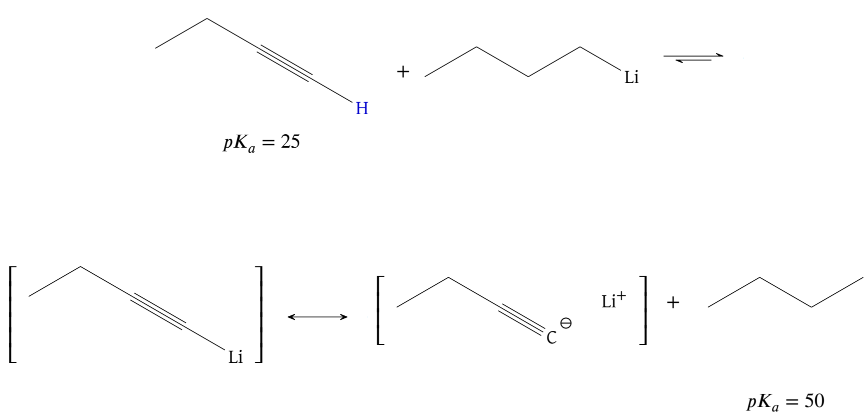


Figure 68: Alkyne acidity and organolithium reagents

2.1 Alkyne Synthesis

Core Concept 2.4: Elimination E2 From Dihaloalkanes

Performing E2 elimination twice on a dihaloalkane bearing adjacent halogen substituents allows for alkyne formation.

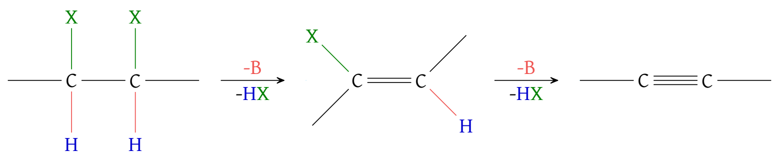


Figure 69: Synthesis of alkyne through double E2 elimination

The example below uses $NaNH_2$ as the base and requires an additional aqueous workup step, owing to the acidity of terminal alkynes (*Core Concept 2.3*) in the presence of sodium amide.

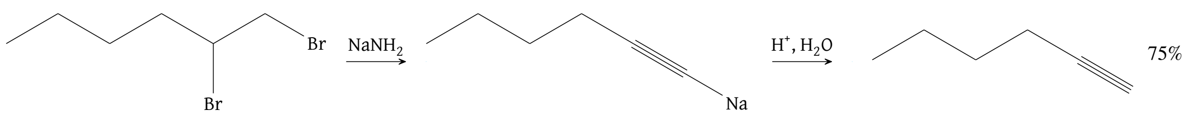


Figure 70: Synthesis of alkyne through double E2 elimination from $NaNH_2$

Core Concept 2.5: Alkylation of Alkynyl Metals

Terminal alkynes can be converted into powerful nucleophiles by deprotonation with strong bases, enabling C–C bond formation through substitution pathways that follow strict S_N2 mechanistic rules.

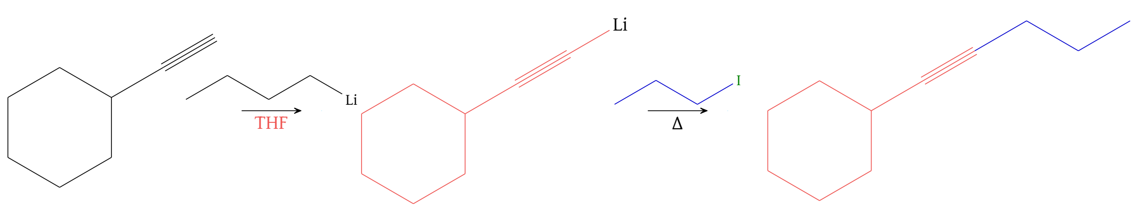


Figure 71: Strong base induced terminal alkyne nucleophilicity

The synthetic utility of alkynyl metals depends on the use of primary ($\boldsymbol{R}_{\text{prim}}$) halides or oxacyclopropanes (*epoxides*) as electrophiles, which minimizes competing E2 elimination side reactions. With asymmetrical epoxides, the acetylide nucleophile attacks regioselectively at the less sterically hindered carbon.

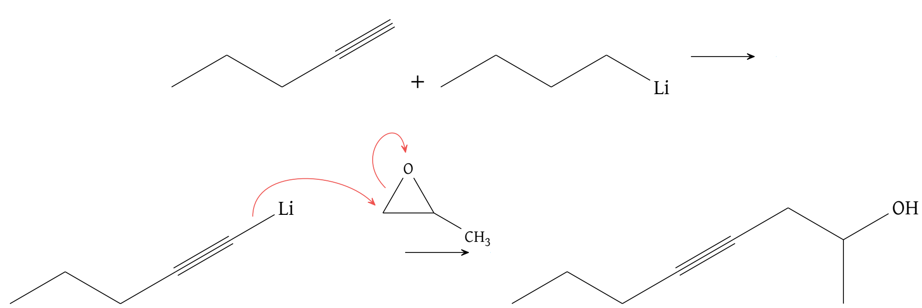


Figure 72: Terminal alkyne nucleophilicity in the presence of an oxacyclopropane

Below are several examples that employ this mechanism.

(a) **Carbonyl Addition with Grignard**

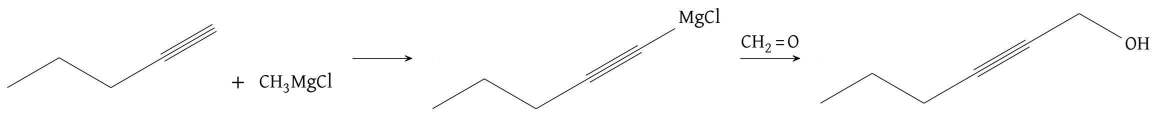


Figure 73: Carbonyl addition with grignard reagent

(b) **Epoxide Opening with Lithium Acetylide**

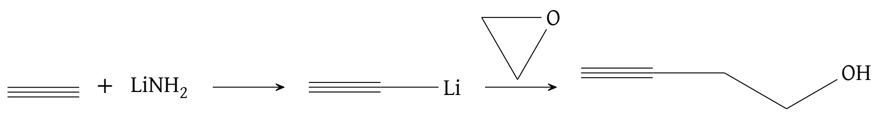


Figure 74: Epoxide opening with lithium acetylide

(c) **Double Grignard Addition of Acetaldehyde**

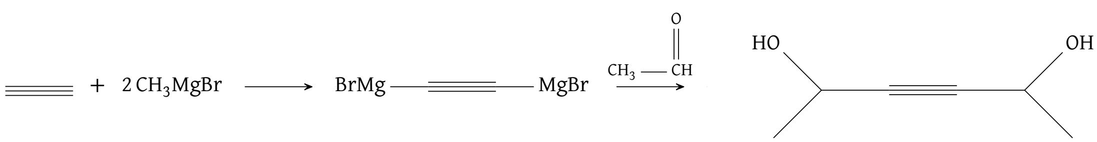


Figure 75: Double grignard addition of acetaldehyde

2.2 Alkyne Reactions

2.2.1 Reductions

Core Concept 2.6: Complete Hydrogenation

Complete hydrogenation of an alkyne triple bond (*the addition of 2* H_2) proceeds analogously to alkene hydrogenation (*Core Concept 1.7*), employing a catalyst such as Pt to lower the reaction's activation energy (E_a). The resulting product is an alkane.

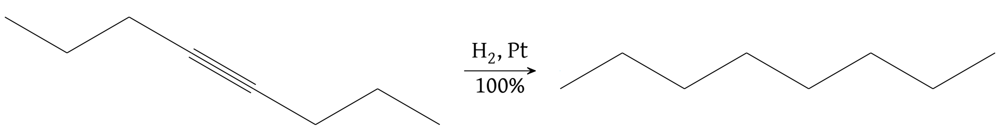


Figure 76: Complete hydrogenation of an alkyne

Core Concept 2.7: Partial Hydrogenation

Partial hydrogenation refers to the addition of only one equivalent of H_2, removing only one of the two π bonds and yielding either a cis or trans product depending on whether the addition is syn or anti, respectively.

- **Lindlar's Catalyst (Syn Addition)**

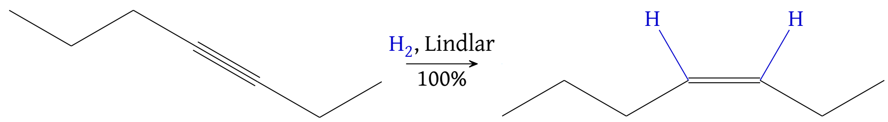


Figure 77: Partial syn hydrogenation of an alkyne with "poisoned" Lindlar's catalyst producing cis-3-heptene

A syn addition, and the resulting cis product, requires the use of a "*poisoned*" **Lindlar's catalyst**.

- **Sodium Reduction (Anti Addition)**

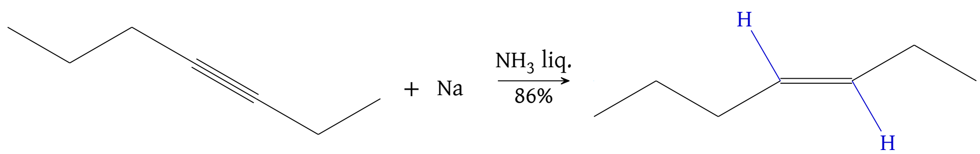


Figure 78: Partial anti hydrogenation of an alkyne with sodium reduction producing trans-3-heptene

The underlying mechanism for this sodium-mediated reduction is shown below.

(1) **First e$^-$ Transfer**

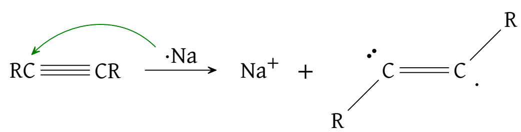


Figure 79: Sodium reduction first e$^-$ transfer; *R groups adopt trans like geometry to minimize steric repulsion*

(2) **First Protonation**

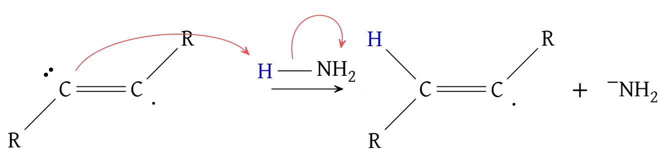


Figure 80: Sodium reduction first protonation

(3) **Second e$^-$ Transfer**

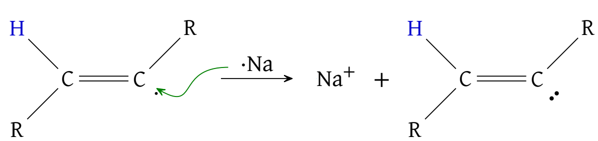


Figure 81: Sodium reduction second e$^-$ transfer

(4) **Second Protonation**

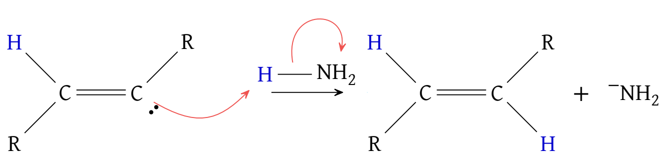


Figure 82: Sodium reduction second protonation

2.2.2 Electrophilic Additions (Markovnikov)

Core Concept 2.8: HX Anti Addition

Addition of **HX** to an alkyne converts the triple bond into a **geminal** (*same-carbon*) dihaloalkane.

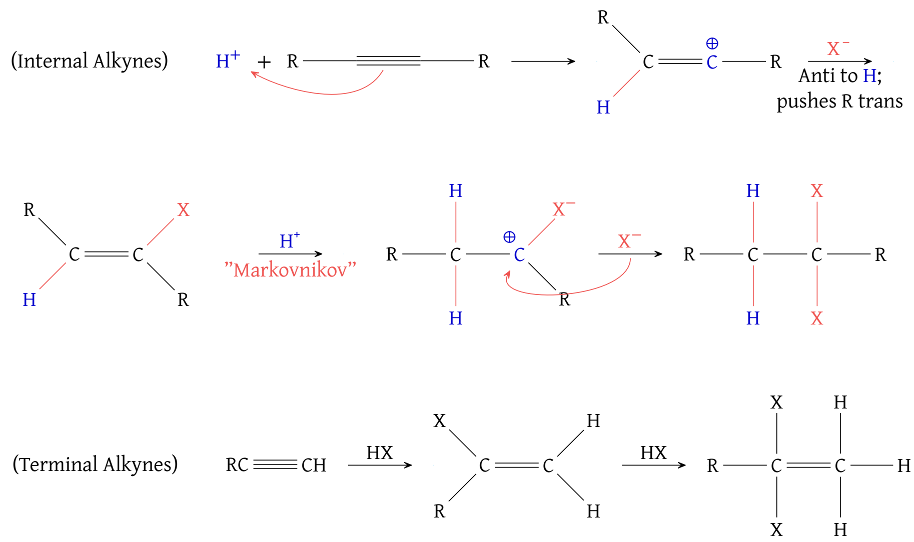


Figure 83: Double Markovnikov HX addition to alkynes resulting in geminal halides

It is important to note that the first HX addition always proceeds in the anti configuration, owing to steric hindrance.

Additionally, although Figure 83 categorizes this HX addition process by internal and terminal alkyne substrates, it is simpler to recognize that the relevant transition states favor carbocation formation at the more substituted carbon (*i.e., Markovnikov's rule*). Because of this tendency, the second HX addition delivers its proton to the same carbon that received the first proton, since the other carbon is already more substituted by the halogen installed in the first addition.

Below are several real reactions illustrating alkyne HX anti additions.

(a) **Symmetrical Internal Alkyne**

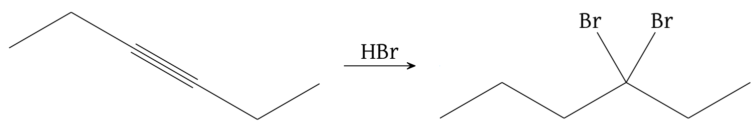


Figure 84: Symmetrical internal alkyne HX anti addition

(b) **Asymmetrical Internal Alkyne**

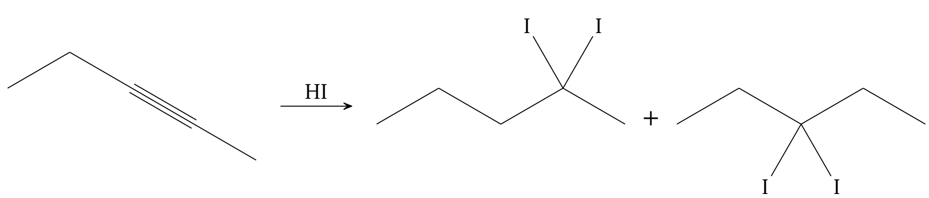


Figure 85: Asymmetrical internal alkyne HX anti addition; *this reaction results in a 50:50 mixture of products*

(c) **Terminal Alkyne**

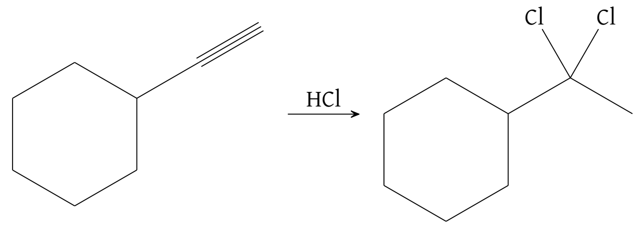


Figure 86: Terminal alkyne HX anti addition

(d) **Synthetic Application (Vicinal to Geminal)**

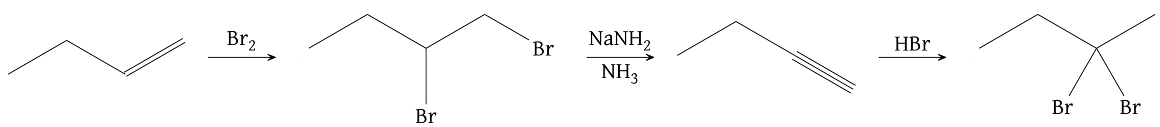


Figure 87: Alkene derived vicinal dihaloalkane to geminal dihaloalkane conversion; *the first step of the above reaction is alkene halogenation, covered in Core Concept 1.11*

Core Concept 2.9: X_2 Anti Addition

Alkyne halogenation **adds X_2** across the triple bond, ultimately installing four halogen atoms (*two on each former triple-bond carbon*) via a transient cyclic halonium intermediate. The mechanism proceeds with an anti addition and is effectively alkene halogenation (*Core Concept 1.11*) occurring twice in succession.

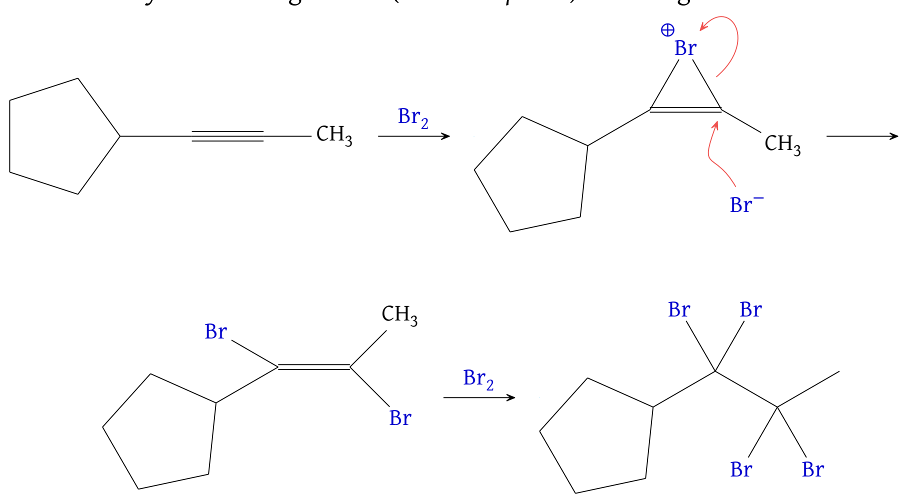


Figure 88: Anti Br_2 addition to an alkyne

Core Concept 2.10: Mercury Catalyzed Markovnikov Hydration

Treating an alkyne with catalytic $HgSO_4$ (*previously seen in Core Concept 1.12*) and H_2O produces an unstable enol intermediate via Markovnikov addition. Subsequent **tautomerization** of this enol, catalyzed by H^+ or ^-OH, yields a ketone regardless of whether the alkyne is terminal or internal (*since the –OH group is installed on the more substituted carbon*).

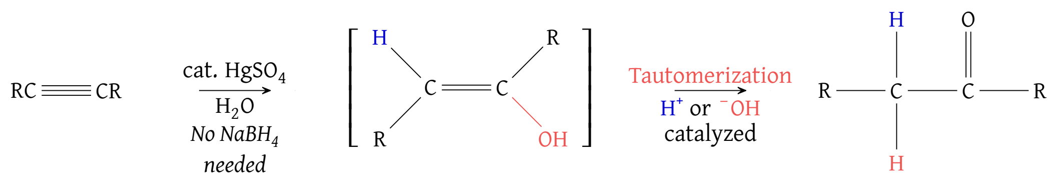


Figure 89: Mercury catalyzed Markovnikov hydration

The underlying mechanism of **tautomerization**—for both the H^+ and ^-OH catalyzed pathways—is shown below.

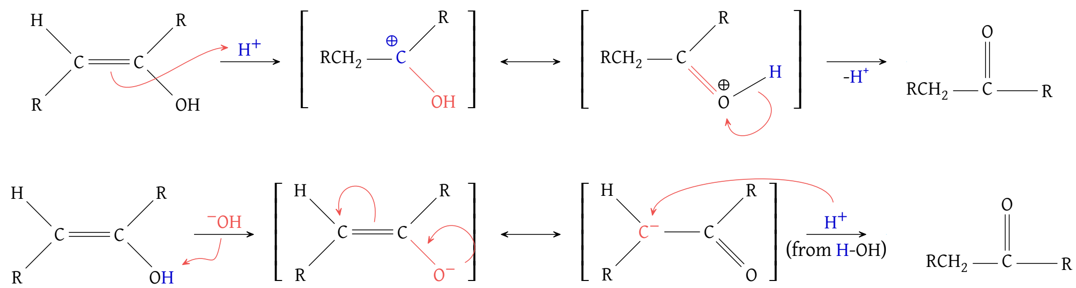


Figure 90: Tautomerization mechanism through H^+ (*top*) and ^-OH (*bottom*)

Below are several practical laboratory applications.

(a) **Asymmetrical Internal Alkyne**

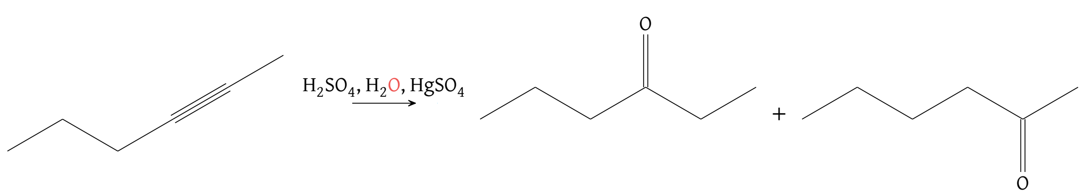


Figure 91: Mercury catalyzed Markovnikov hydration on an asymmetrical internal alkyne; *both products are formed at a 50:50 ratio*

(b) **Terminal Alkyne**

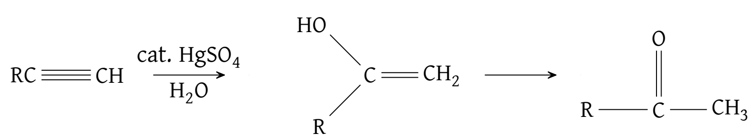


Figure 92: Mercury catalyzed Markovnikov hydration on a terminal alkyne; *reaction results in a methyl ketone*

Core Concept 2.11: Radical Hydrobromination

In alkynes, **radical hydrobromination** is the **single addition of HBr** via a peroxide-generated (ROOR) radical mechanism. The reaction follows the same mechanism described for the alkene version of this electrophilic addition (*Core Concept 1.19*), but yields a mixture of products owing to free rotation about the newly formed double bond.

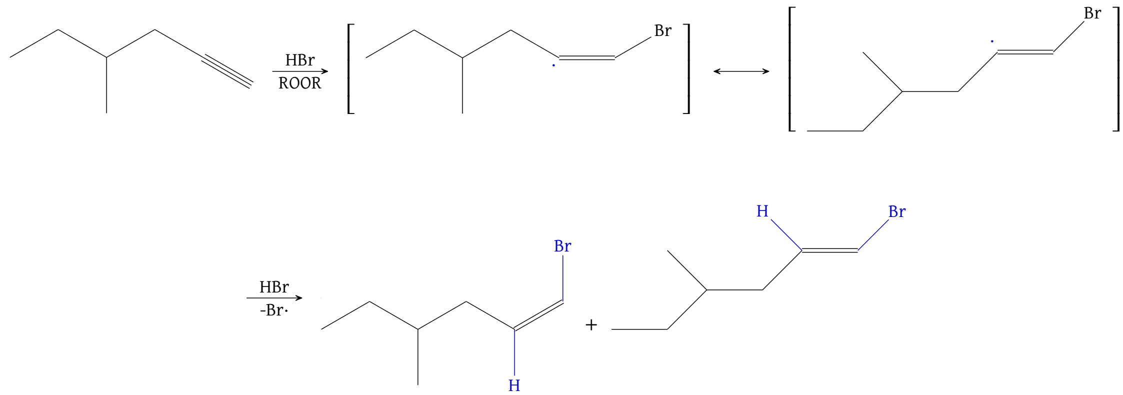


Figure 93: Radical hydrobromination

Unlike the additions seen in HX anti addition (*Core Concept 2.8*) and X_2 anti addition (*Core Concept 2.9*), this reaction does not necessarily proceed through a second iteration to further convert the resulting alkene into an alkane. This is because the mechanism requires an electron-rich π bond, a condition the resulting bromoalkene fails to meet: the highly electronegative bromine substituent withdraws e^- density from the double bond (*a condition that does not affect an unsubstituted alkene*). Consequently, because the starting alkyne is considerably more reactive than the bromoalkene product, the radicals preferentially consume all remaining alkyne before reacting further with the bromoalkene intermediate.

If, however, alkynes are exposed to an excess of HBr and peroxides, a second radical addition can occur, governed by the same regiochemical stability considerations, to form a **geminal dihaloalkane**—the same product obtained from HX anti addition (*Core Concept 2.8*). This makes the additional peroxide unnecessary if the original goal was simply to form a geminal dihaloalkane.

Core Concept 2.12: Hydroboration-Oxidation

Alkyne **hydroboration-oxidation** employs a bulky borane such as R_2BH (*where R is a bulky group, e.g., cyclohexyl*) to ensure that only one π bond undergoes hydroboration. *Recall that alkene hydroboration-oxidation (Core Concept 1.13) simply uses BH_3, since there is only a single π bond to begin with.* The mechanism otherwise follows the same process as the alkene version of this reaction (*Core Concept 1.13*) and effectively achieves an anti-Markovnikov hydration.

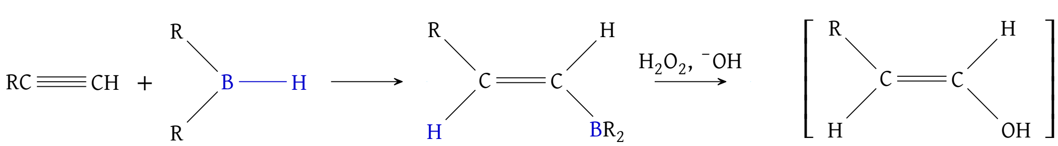


Figure 94: Hydroboration-oxidation

The resulting enol can then be carried through the previously covered tautomerization process to be converted into an aldehyde.

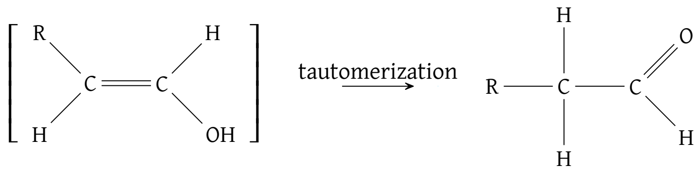


Figure 95: Tautomerization

The figure below offers a direct comparison with **mercury-catalyzed Markovnikov hydration** (*Core Concept 2.10*), which proceeds similarly but yields a ketone rather than an aldehyde.

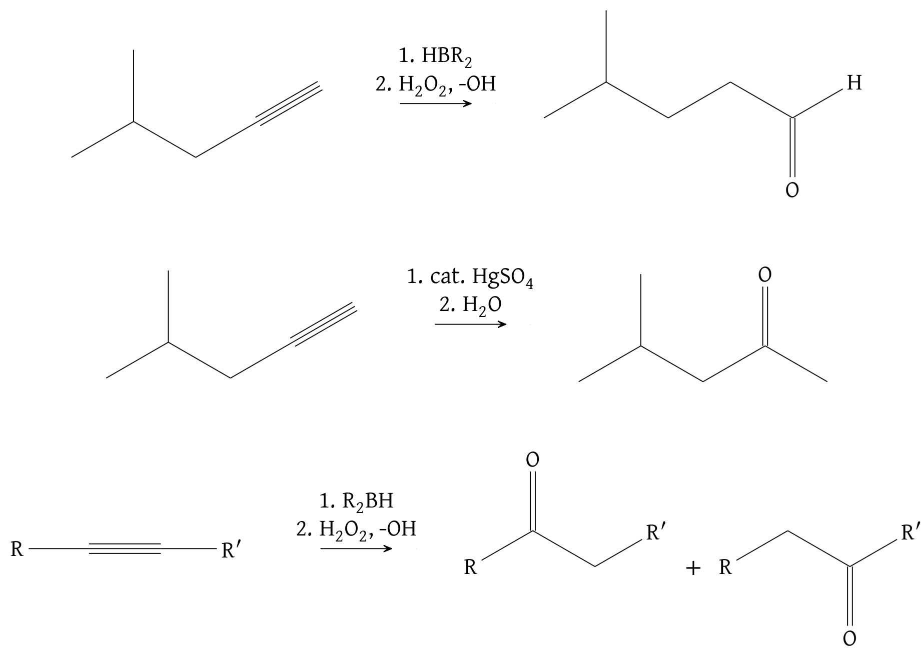


Figure 96: Hydroboration-oxidation (*top, bottom*) vs mercury catalyzed Markovnikov hydration (*middle*); *internal alkynes (bottom) result in a mixture of products*